

Production Quality Management Digitization

Final Implementation Proposal

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1. Executive Summary

NPIC's current production and quality inspection processes rely on paper records, resulting in three core problems: delayed QC results, absence of decision-making support during night shifts, and information asymmetry between departments.

This project will build a dedicated QC Real-Time Management System (Phase 1) for NPIC, covering the complete workflow from production batch creation to inspection result decisions and qualified product release to the packaging queue — replacing all existing paper records.

System development is complete. Both parties have concluded the final requirements meeting and are now entering the testing and go-live phase. This document details the system scope, testing plan, implementation plan, and investment.

2. Current State Diagnosis

2.1 QC Information Is Significantly Delayed

All tests can only be completed near the end of each shift; results are not available in real time. This means:

- An information gap of several hours between when a cart finishes drying and when results are available;
- Quality issues cannot be addressed in real time; problem batches may continue to flow downstream;
- Drying room check-in records depend on manual paper forms, causing delays and errors.

2.2 Night Shift Decisions Lack Real-Time Support

Because test results are only summarized at end of shift, management cannot remotely determine during the night shift which batches are qualified and which require disposition.

2.3 Cross-Department Information Silos

The warehouse and packaging teams lack shared real-time records. Batch flow status exists only in each department's paper records or verbal communication. When disputes arise, there is no evidence to consult.

2.4 Drying Room Check-In/Out Relies on Manual Paper

Cart number entries are prone to duplication errors. Every morning shift, staff must manually transcribe check-in forms for all pending drying carts — taking over 3 hours — with frequent data errors.

2.5 Sales Team Cannot Access Real-Time Production Status

Because the various systems are not connected, the sales team cannot monitor production progress and inventory status in real time, making delivery commitments and inventory planning unreliable.

3. System Scope: QC Real-Time Management System

3.1 System Workflow

The system covers the complete workflow:

- Work order creation → Label printing → Production staff attach labels to carts
- Barcode scanner scan → Confirm cart entry into drying queue
- Manual drying room assignment → Automatic countdown timer starts
- Expiry alert → Batch checkout → Sampling notification
- Lab technician enters water activity reading → System auto-judges Pass / Fail
- Qualified products enter packaging queue → Management releases with one click
- Non-conforming products trigger disposition workflow (Re-test / Re-dry / Scrap)
- Packaging dept views all released carts, multi-selects, batch dispatches

3.2 Core Functional Modules

Module A — Production Batch & Cart Management

- Work order creation generates production batches; system auto-generates cart numbers (WO#-001 through WO#-N)
- Each cart label is bound to its system record; scan to view real-time status
- Work orders can be printed; production staff attach labels to corresponding carts

Module B — Drying Room Real-Time Monitoring

- Real-time display of carts currently in each drying room and their status
- Automatic calculation of estimated completion time based on configured target drying duration
- Automatic expiry alerts to prevent over-drying or failure to retrieve on time

Module C — Sampling & Inspection Workflow

- Configurable sampling rules to reduce unnecessary inspections
- Lab technicians enter water activity readings directly; system auto-judges Pass/Fail
- Inspection results reported in real time — no need to wait for end-of-shift roll-up

Module D — Non-Conforming Product Disposition Workflow

- Failed batches immediately trigger a disposition workflow: Re-test / Re-dry / Ambient dry / Scrap
- Disposition decisions are fully recorded with a complete audit trail

Module E — Management Dashboard

- Real-time overview: batches currently drying, pending inspection, today's pass rate, batches requiring attention
- Qualified batches released to packaging queue with a single click

Module F — Complete Batch Traceability

- Any batch is fully traceable: Production batch → Sub-lot (cart) → Sampling record → Inspection result → Disposition → Release time
- Provides a complete evidence chain when inter-departmental disputes arise

Module G — Product & Inspection Standard Management

- Create products and customize basic information
- Define inspection items and pass/fail ranges per product (dual-limit or upper-limit-only modes)
- Configurable manual-review zone: borderline results escalated to a supervisor

Module H — Packaging Release Queue (New)

- Wendy (Packaging Dept Leader) logs in to view a complete list of all QC-released carts
- Supports multi-select and batch Dispatch — confirm multiple carts in one action
- Release records are linked to corresponding work orders and inspection results; packaging and warehouse data stay in sync in real time

3.3 Problems Solved

Current Problem	System Solution
Test results only summarized at end of shift — no real-time decisions	Lab enters results on-site; outcome visible immediately
Packaging team cannot anticipate incoming qualified inventory	Qualified products enter packaging queue instantly after release
Problem batches discovered too late; cannot be resolved within the shift	Failed batches trigger disposition workflow in real time, same shift
Drying room check-in/out entirely manual — time-consuming and error-prone	Barcode scanner check-in/out; system monitors drying status and sends automatic expiry alerts
Packaging team has no view of released carts; batch operations are inefficient	Module H provides complete release queue with multi-select batch Dispatch

4. Technical Architecture

Aspect	Details
Devices	Accessible on Windows / Mac computers and tablets; responsive interface adapts to screen size
Data Storage	Cloud database with real-time backups; auto-syncs when connectivity is restored after outages
Data Security	Enterprise-grade encrypted storage; data belongs to NPIC; no third-party access
Access Control	Role-based permissions; each position only sees and operates within its authorized scope

Aspect	Details
Deployment & Maintenance	Deployed and maintained by Crave & Cook; NPIC requires no server infrastructure or IT staff

5. Deliverables & Provided Equipment

Crave & Cook will deliver the following to NPIC as part of this project:

Software

- QC Real-Time Management System (fully developed and configured per NPIC requirements)
- User accounts and role-based permission setup
- Role-specific operation manuals (print and digital)

Hardware (delivered with the project)

- 2 barcode scanners (for drying room cart check-in / check-out)
- 3 date stamps
- 2 ink pads

6. Testing Plan

Testing is conducted using 2 real work orders that exercise the complete system workflow. IT must complete account setup and permission assignment before testing begins.

6.1 Pre-Test Setup (Before Day 1)

Task	Responsible	Details
Create account for Du Gong; assign drying room operator permissions	IT	Can scan check-in/checkout, assign drying rooms, batch checkout
Create account for Vivian; assign inspection module permissions	IT	Can receive sampling notifications, enter water activity results, handle Pass/Fail
Create account for Wendy; assign packaging module permissions	IT	Can view released cart list, multi-select, batch dispatch

Task	Responsible	Details
Create accounts for production printing staff	IT	Can create work orders, print labels
Verify barcode scanner connects to system correctly	Crave & Cook + Du Gong	Drying room device fully tested and operational

6.2 Testing Timeline (5 Working Days)

Day	Phase	Responsible	Before	After
Day 1	Work Order Creation & Printing	Production Supervisor	Each cart had one label with two barcodes on it	Consolidated into a single QR code label; production staff print and attach to each cart; scan binds the cart to its work order
Day 2-3	Production	Production Staff	No change from existing process	
Day 3-4	Drying Room Entry	Du Gong	Manual paper form for room entry; handwritten date	Barcode scanner scan completes system check-in; date stamp for check-in/out date; system auto-starts drying timer and records room assignment
Day 4-5	Drying Room Exit	Du Gong	No system record; no defined exit timestamp	Batch checkout in system; exit time automatically recorded
Day 4-5	Inspection & Testing	Vivian + Release Approver	End-of-shift summary handed off as Excel; no real-time visibility or same-shift decisions	Each completed test entered immediately; results visible in real time; disposition decisions made within the same shift
Day 4-5	Packaging Dispatch	Wendy	Only physical Pass/Fail stamps on the floor to distinguish packable batches; relied on manual on-site communication	Real-time view of all released carts in system; multi-select batch Dispatch

6.3 Acceptance Criteria

- Both work orders successfully complete the full cycle — from creation to dispatch — with accurate data and zero recording errors
- All user accounts have correct permissions; each role can only access its authorized functions
- All notifications delivered to correct personnel at each step (sampling notifications, expiry alerts, release notifications)
- Barcode scanner check-in/out, batch checkout, and batch dispatch functions work correctly
- System data matches actual operations with no missing records

Total testing duration: approximately 5 working days (including production cycle).

7. Implementation Plan

Implementation proceeds in four phases, from data initialization to full go-live.

Phase 1: Data Entry & System Initialization

Task	Responsible	Details
Product catalog entry (SKU information, inspection limits)	Crave & Cook + NPIC QC Manager	Enter all product basic info and water activity pass/fail ranges
Inspection template configuration	Crave & Cook	Configure Pass/Fail judgment rules and manual-review zones per SKU
Work order format confirmation and configuration	Crave & Cook + Production Supervisor	Confirm WO number format, configure print template
Account creation and permission assignment	IT + Crave & Cook	Create all user accounts, assign role-based permissions

Phase 2: On-Site Installation & Device Setup

Role / Device	Tasks	Responsible
Drying Room (Du Gong)	Install and test barcode scanner and tablet; verify check-in/out and room assignment functions	Crave & Cook + Du Gong

Role / Device	Tasks	Responsible
Testing Lab (Vivian)	Install and test tablet/PC; verify sampling notification receipt and result entry functions	Crave & Cook + Vivian
Packaging Dept (Wendy)	Install and test tablet; verify released cart list and batch dispatch functions	Crave & Cook + Wendy
Management Dashboard	Management can access real-time full-system overview on PC or tablet	Crave & Cook

Phase 3: Parallel Run

- Paper records and system run in parallel for 1 week to validate data accuracy
- Issues identified during the parallel run are resolved
- Client confirms system meets acceptance criteria and signs the Acceptance Certificate

Phase 4: Role-Based Training & Go-Live

Trainee	Training Content	Duration	Trainer
Printing Staff	Work order creation, label printing, cart numbering rules	1 hour	Crave & Cook
Drying Room Operator	Barcode scanner check-in/out, drying room assignment, batch checkout, expiry alert handling	2 hours	Crave & Cook
Testing Lab Leader	Sampling notification receipt, water activity entry, Pass/Fail disposition workflow	2 hours	Crave & Cook
Packaging Dept Leader	View released cart list, multi-select, batch dispatch operation	1 hour	Crave & Cook

After each training session, role-specific operation manuals are provided (both print and digital). Upon completion of all training, the system fully cuts over to digital; paper processes are discontinued; official go-live.

8. Expected Benefits

Operational Efficiency Gains

- Drying room check-in/out fully scan-based, replacing repeated manual writing — estimated 50%+ reduction in related labor time

- System automatically tracks and recommends sample batches, eliminating the lab's manual counting and sample selection process
- Inspection data entered directly in the system, replacing write-then-transcribe-to-Excel double-entry — estimated 30%+ reduction in related labor hours
- Inspection results visible in real time — decision lead time reduced from end-of-shift report to instant visibility

Quality Control Strengthened

- Non-conforming batch alerts pushed in real time; management completes disposition within the same shift
- Inspection anomalies synced to the production line in real time, enabling rapid process adjustments
- Complete inspection history accumulated for long-term quality trend analysis and continuous improvement

Cross-Department Collaboration

- QC and packaging teams share a single real-time data platform, eliminating the information black hole
- Full batch traceability ensures inter-departmental disputes are backed by verifiable records
- Reduced communication overhead and departmental friction caused by information asymmetry

9. Investment

Billing reference rate: USD \$100 / hour. Phase 1 is quoted as a fixed project fee for the scope below.

Cost Component	Hours	Subtotal (USD)
Requirements & Process Mapping (SKU standards, WO format, permission planning)	20 hrs	\$2,000
System Configuration & Data Initialization (product catalog, inspection templates, account & permission setup)	52 hrs	\$5,200
Core Modules Development & Customization (Modules A–G)	150 hrs	\$15,000
Module H — Packaging Release Queue (new development)	12 hrs	\$1,200
Cloud Deployment, Domain Setup & Data Migration	12 hrs	\$1,200

Cost Component	Hours	Subtotal (USD)
On-Site Installation, Testing & Acceptance Support (parallel run, issue resolution)	18 hrs	\$1,800
Go-Live Training & Documentation (per-role training, operation manuals)	15 hrs	\$1,500
Project Management & Coordination	9 hrs	\$900
Hardware & Equipment (2 barcode scanners, 3 date stamps, 2 ink pads)	—	\$500
Total	288 hrs	\$29,300

Add-On Option: IP Access Control

This section describes an optional add-on service, not included in the Phase 1 base price. It may be purchased separately as needed.

Overview. Using Vercel Edge Middleware, access to the system is restricted to a whitelist of approved IP addresses. Requests from any IP not on the whitelist are blocked at the edge before the page loads — this cannot be bypassed by end users.

Use Cases

- NPIC's factory network has a static IP — add it to the whitelist directly
- If no static IP is available, a VPN can be configured for relevant staff; all access routes through the VPN's fixed exit IP, which is then whitelisted
- Multiple IPs can be whitelisted (e.g., factory, office, management home) — NPIC IT manages additions and removals via the Vercel dashboard

Deliverables

- Vercel Edge Middleware IP whitelist implementation (including test verification)
- IP management documentation for NPIC IT self-service maintenance

Item	Amount (USD)	Notes
IP Access Control (Add-On)	\$2,000	One-time fee; purchased separately; not included in Phase 1 pricing

10. About Us

We specialize in delivering tailored digital transformation solutions for manufacturing clients, with particular expertise in translating complex production workflows into intuitive, easy-to-use management systems. The QC system designed for NPIC is purpose-built for the actual workflows of the food manufacturing industry — not a generic ERP retrofitted to the use case.

Thank you for the trust placed in Crave & Cook, Inc. We look forward to partnering with NPIC to drive its digital transformation in production management.

This document is a confidential commercial proposal intended solely for NPIC's internal decision-making.